Emily Kuo
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EDUCATION

M.S. in Mechanical Engineering University of California, San Diego Expected June 2028
B.S. in Bioengineering University of California, Riverside Sept 2022 - June 2026

RESEARCH EXPERIENCE

Infant Model For Testing of Assistive Devices PRT lab, UCR Oct 2023 - June 2026

- Led a four-person team in the design, development, fabrication, and validation of a physical testing platform for infant assistive devices.
- Designed and fabricated SolidWorks CAD components and assemblies for a biomechanical testing system.
- Evaluated the model's center of mass and range of motion compared with published anthropometric values.
- First author of a published manuscript in Data in Brief.

Classification of Infant Locomotion PRT lab, UCR Apr 2023 - June 2026

- Analyzed over 400 videos of infant movement using OpenPose and Kinovea to evaluate infant kinematic patterns.
- Utilized OpenPose for automated pose estimation and Kinovea for video-based motion analysis.
- Contributed to the development of the KIRA dataset, documenting full-body infant kinematics during sitting and reaching activities in natural environments.

Infant Wearable Exoskeleton Suit PRT lab, UCR Apr 2023 - May 2023

- Lead a three person team in fabricating over 100 pneumatic actuator components for an infant wearable exosuit.
- Contributed to the fabrication and assembly of pneumatic actuators intended for integration into a wearable assistive device.
- Worked with alternative actuator configurations to investigate differences in expansion and actuation behavior.

PROJECTS

Microfluidic Device Senior Design, UCR Sept 2025 - June 2026

- Designed and fabricated a microfluidic device for producing hollow-core BIPORES fibers used in the development of artificial scaffolds for tissue engineering.
- Co-designed device components and assembly procedures to establish a functional and repeatable fabrication system.
- Worked across mechanical and chemical engineering disciplines to evaluate device performance and fabrication compatibility.
- Performed statistical analyses to evaluate device functionality and validate experimental results.

Modified ENDO Continuum Robot Surgical Arm Design of Robots, UCR Apr 2026 - May 2026

- Modified an existing ENDO continuum robot surgical arm design, reducing the original three-segment configuration to a two-segment architecture.
- Integrated servo motors and tendon-driven actuation components to enable controlled bending of the continuum robot.
- Conducted repeated servo actuation trials to characterize motor response and assess consistency across actuation channels.

Pulmonary Hemorrhage Stents Circulation Physiology, UCR Jan 2025 - Mar 2025

- Developed three-dimensional stent geometries in SolidWorks to investigate the effects of stent design on pulmonary blood flow.
- Investigated the relationship between stent geometry and resulting flow efficiency through computational analysis.

SKILLS

Mechanical Design: FDM 3D printing, Resin 3D printing, Laser cutting, Soldering
Modeling & Simulations: COMSOL multiphysics, SOLIDWORKS
Programming & Data Analysis: MATLAB, Python, GraphPad Prism
Experimental & Research Methods: Device prototyping & validation, Statistical analysis, Technical documentation and scientific writing

PUBLICATIONS

Kuo, E., Macaluso, C., Rauch, C., Gowda, A., Ortega, J., Castillo, M.C., Sahin, I., and Kokkoni, E. Design and Manufacturing Data for an Infant Upper Body Physical Model for Bench Testing of Assistive Devices (published, August 2026).

Sahin, I., McCarty, T., Kouvoutsakis, G., **Kuo, E.**, Macaluso, C., Gowda, A., Castillo, M., Ortega, J., Mestas, J., Modi, A., Barrundia, D., Chidambaram, K., Nguyen, L., Peiris, P., Dechemi, A., Roy-Chowdhury, A.K., Karydis, K., and Kokkoni, E. KIRA: A Dataset of Full-body Kinematics of Infants Sitting and Reaching in Natural Settings. *Scientific Data* (submitted, Sept 2026).